

June 22, 2026

Dr. Yufei Han
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Rennes, France

Dear Dr. Yufei Han,

I am writing to formally submit my application for the PhD position focusing on using Large Language Models to repair vulnerabilities in repository-level software projects (SecLLM4SVD). With a State Engineer degree in AI & Data Science and extensive experience bridging complex algorithms with real-world infrastructure, I am highly motivated to tackle the security vulnerabilities of LLMs in code analysis.

In my recent engineering theses, I focused heavily on ensuring algorithmic reliability in mission-critical environments. For my Master's thesis on the "Cardiovascular Medical Digital Twin," I engineered MedicalMambaTS, leveraging linear-time state-space models for sub-millisecond inference. Crucially, I also developed PI-DeepONet, a Physics-Informed Operator Network that explicitly maps physical constraints (0D Windkessel equations) into the neural architecture via PyTorch Autograd. This experience of embedding structural, deterministic rules into a "black-box" neural network directly translates to the goals of your WP2 and WP3.

To prevent LLMs from being misled by adversarial attacks (like variable renaming or dead code insertion) during vulnerability detection, we cannot rely purely on associative token prediction. Instead, I propose using the structural logic of Control Flow Graphs (CFG) and Data Flow Graphs (DFG) to explicitly constrain the LLM's reasoning paths via reinforcement learning, acting as a deterministic "physics-informed" loss function for code semantics.

I am highly proficient in Python, PyTorch, and navigating complex codebases, and I am eager to apply my background in structured deep learning to software security within the PIRAT team.

Thank you for your consideration. I look forward to the possibility of discussing this research further.

Sincerely,
Ismail Aissaoui
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